

Easy Java Assignments

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1. Create a class with a method that prints "This is parent class" and its subclass with another method that prints "This is child class". Now, create an object for each of the class and call

- 1 - method of parent class by object of parent class
- 2 - method of child class by object of child class
- 3 - method of parent class by object of child class

```
4  public class Parent1 {
5      void DisplayParent() {
6          System.out.println("This is Parent Class");
7      }
8  }
9  class Child2 extends Parent1{
10     void DisplayChild() {
11         System.out.println("This is Child Class");
12     }
13 }
14 class ChildParent{
15     public static void main(String[] args){
16         Parent1 p = new Parent1();
17         p.DisplayParent();
18         Child2 c = new Child2();
19         c.DisplayChild();
20         c.DisplayParent();
21     }
22 }
23
```

2. Create a class named 'Member' having the following members:

Data members

- 1 - Name
- 2 - Age
- 3 - Phone number
- 4 - Address
- 5 - Salary

It also has a method named 'printSalary' which prints the salary of the members.

Create an object of the above class and take input from the console for each of the fields and finally display the data.

Now create a class inherited from Members called PrimeMembers and add the properties

1. JoiningYear
2. JoiningFees
3. isActive

To the PrimeMembers add a method called display and display all the fields from the parent class and derived class. Each of the above classes should have their respective accessors and mutators.

```
import java.io.*;
public class Member {
    private String name;
    private int age;
    private String phoneNumber;
    private String address;
    private double salary;

    // (Getters)
    public String getName() { return name; }
    public int getAge() { return age; }
    public String getPhoneNumber() { return phoneNumber; }
    public String getAddress() { return address; }
    public double getSalary() { return salary; }

    // Mutators (Setters)
    public void setName(String name) { this.name = name; }
    public void setAge(int age) { this.age = age; }
    public void setPhoneNumber(String phoneNumber) { this.phoneNumber =
phoneNumber; }
    public void setAddress(String address) { this.address = address; }
    public void setSalary(double salary) { this.salary = salary; }

    // Method to print salary
    public void printSalary() {
        System.out.println("Salary: $" + salary);
    }
}

// Derived class
class PrimeMembers extends Member {
    private int joiningYear;
    private double joiningFees;
    private boolean isActive;

    //Getters
    public int getJoiningYear() { return joiningYear; }
    public double getJoiningFees() { return joiningFees; }
    public boolean getIsActive() { return isActive; }

    //Setters
    public void setJoiningYear(int joiningYear) { this.joiningYear = joiningYear;
}
    public void setJoiningFees(double joiningFees) { this.joiningFees =
joiningFees; }
    public void setIsActive(boolean isActive) { this.isActive = isActive; }

    // Method to display all fields
```

```

    public void display() {
        System.out.println("\n--- Member Details ---");
        System.out.println("Name: " + getName());
        System.out.println("Age: " + getAge());
        System.out.println("Phone Number: " + getPhoneNumber());
        System.out.println("Address: " + getAddress());
        printSalary();
        System.out.println("Joining Year: " + joiningYear);
        System.out.println("Joining Fees: $" + joiningFees);
        System.out.println("Status: " + (isActive ? "Active" : "Inactive"));
    }
}

class consoleInput {
    public static String getString() {
        try {
            byte arrInput[] = new byte[100];
            int length = System.in.read(arrInput);
            if (length <= 0) return "";
            // '\r\n' (Enter key) ko handle karne ke liye
            int count = (length >= 2 && arrInput[length - 2] == '\r') ? length - 2
: length - 1;
            if (count < 0) count = 0;
            byte[] arrFinal = new byte[count];
            System.arraycopy(arrInput, 0, arrFinal, 0, count);
            return new String(arrFinal);
        } catch (Exception e) {
            e.printStackTrace();
            return null;
        }
    }

    public static int getInt() {
        return Integer.parseInt(getString().trim());
    }

    public static float getFloat() {
        return Float.parseFloat(getString().trim());
    }

    public static double getDouble() {
        return Double.parseDouble(getString().trim());
    }

    public static boolean getBoolean() {
        return Boolean.parseBoolean(getString().trim());
    }
}

class UseGetSet {
    public static void main(String[] args) {
        PrimeMembers pm = new PrimeMembers();

        // Custom consoleInput helper class se input le rahe hain
        System.out.print("Enter Name: ");
        pm.setName(consoleInput.getString());
    }
}

```

```

        System.out.print("Enter Age: ");
        pm.setAge(consoleInput.getInt());

        System.out.print("Enter Phone Number: ");
        pm.setPhoneNumber(consoleInput.getString());

        System.out.print("Enter Address: ");
        pm.setAddress(consoleInput.getString());

        System.out.print("Enter Salary: ");
        pm.setSalary(consoleInput.getDouble());

        System.out.print("Enter Joining Year: ");
        pm.setJoiningYear(consoleInput.getInt());

        System.out.print("Enter Joining Fees: ");
        pm.setJoiningFees(consoleInput.getDouble());

        System.out.print("Is Active? (true/false): ");
        pm.setIsActive(consoleInput.getBoolean());

        // Displaying all details
        pm.display();
    }
}

```

Input

```

enter Name: anushri patil
Enter Age: 23
Enter Phone Number: 67585330223
Enter Address: pune
Enter Salary: 1000000
Enter Joining Year: 1000000
Enter Joining Fees: 100000
Is Active? (true/false): true

```

Output

```

--- Member Details ---
Name: anushri patil
Age: 23
Phone Number: 67585330223
Address: pune
Salary: $1000000.0
Joining Year: 1000000
Joining Fees: $100000.0
Status: Active

```

1. Create a class named 'Rectangle' with two data members 'length' and 'breadth' and two methods to print the area and perimeter of the rectangle respectively. Its constructor having parameters for length and breadth is used to initialize length and breadth of the rectangle. Let class 'Square' inherit the

'Rectangle' class with its constructor having a parameter for its side. Print the area and perimeter of a rectangle and a square.

```
2.      public class Q3Rectangle {
3.      int length;
4.      int breadth;
5.      Q3Rectangle(int l,int b){
6.      length = l;
7.      breadth = b;
8.      }
9.      void area() {
10.     System.out.println("Rectangle Area:" + (length*breadth));
11.     }
12.     void perimeter() {
13.     System.out.println("Rectangle Perimeter:" +
14.     (2*(length+breadth)));
15.     }
16.     class Square extends Q3Rectangle{
17.     Square(int side){
18.     super(side,side);
19.     }
20.     //method overriding
21.     @Override
22.     void area() {
23.     System.out.println("Square Area:" + (length*breadth));
24.     }
25.     @Override
26.     void perimeter() {
27.     System.out.println("Square Perimeter:" +
28.     (2*(length+breadth)));
29.     }
30.     }
31.     class Main {
32.     public static void main(String [] args) {
33.     Q3Rectangle r = new Q3Rectangle(10,5);
34.
```

```

35.         System.out.println("Rectangle");
36.         r.area();
37.         r.perimeter();
38.
39.         Square s = new Square(5);
40.         System.out.println("Square");
41.
42.         s.area();
43.         s.perimeter();
44.     }
45. }
46.
47.

```

•

48.

49. Create a class named 'Shape' with a method to print "This is shape". Then create two other classes named 'Rectangle', 'Circle' inheriting the Shape class, both having a method to print "This is rectangular shape" and "This is circular shape" respectively. Create a subclass 'Square' of 'Rectangle' having a method to print "Square is a rectangle". Now call the method of 'Shape' and 'Rectangle' class by the object of 'Square' class.

```

50.         class ShapeBase {
51.         void shapeDisplay() {
52.         System.out.println("This Is Shape");
53.         }
54.         }
55.         class Rect extends ShapeBase{
56.         void rectangleDisplay() {
57.         System.out.println("This is REctangle");
58.         }
59.         }
60.         class circuler extends ShapeBase{
61.         void circleDi() {
62.         System.out.println("This is shape");
63.         }
64.         }
65.         class Square2 extends Rect {

```

```
66.         void display() {
67.             System.out.println("This is Square");
68.         }
69.     }
70.     public class Shape2 {
71.         public static void main(String [] args) {
72.
73.             Square2 s = new Square2();
74.             s.shapeDisplay();
75.             s.rectangleDisplay();
76.             s.display();
77.         }
78.     } 79.
```